

Evidence of Problem Research

A Forensic Audit of Prescription Safety in Bangladesh

Prepared for the GRIC Startup League. Project: IntelliScript BD, a sovereign health identity and prescription safety network designed for power-unstable, high-risk

EXECUTIVE CERTIFICATION

This dossier certifies that the N=22 Bangladesh Digital Health & Safety Survey dataset and the field evidence package were gathered and compiled during April-May 2026. Survey worksheet timestamps verify responses between 03 May and 06 May 2026. Field photos are presented as founder-supplied primary evidence for problem validation.

N=22

PRIMARY SURVEY

Bangladesh Digital Health & Safety Survey; May 2026.

12

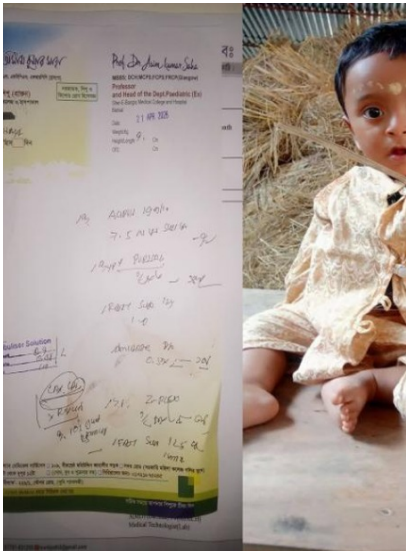
PAGE AUDIT

Problem research dossier with source register.

2026

FIELD WINDOW

April-May primary field and survey evidence.



Anchor Evidence: handwritten prescription hazard and patient-side documentation.

STATEMENT OF AUTHENTICITY

H.O.L.M.E.S. Research Organization confirms that this document is a synthesis of raw survey data, founder field photographs, legal/regulatory anchors, and peer-reviewed/public health evidence. It is structured as an evidence dossier, not as a clinical diagnosis of any individual appearing in field photographs. Patient faces and clinical contexts are used only as problem-validation evidence supplied by the founders for competition submission.

The Quantitative Crisis: 96.75% Outpatient Diagnosis Gap

Rahman et al. reviewed 400 outpatient prescriptions from a secondary care district hospital in Pabna, Bangladesh. The critical finding: diagnosis was missing from 96.75% of prescriptions.

96.75%

DIAGNOSIS NOT MENTIONED

Proxy for the outpatient clinical-context vacuum.

400

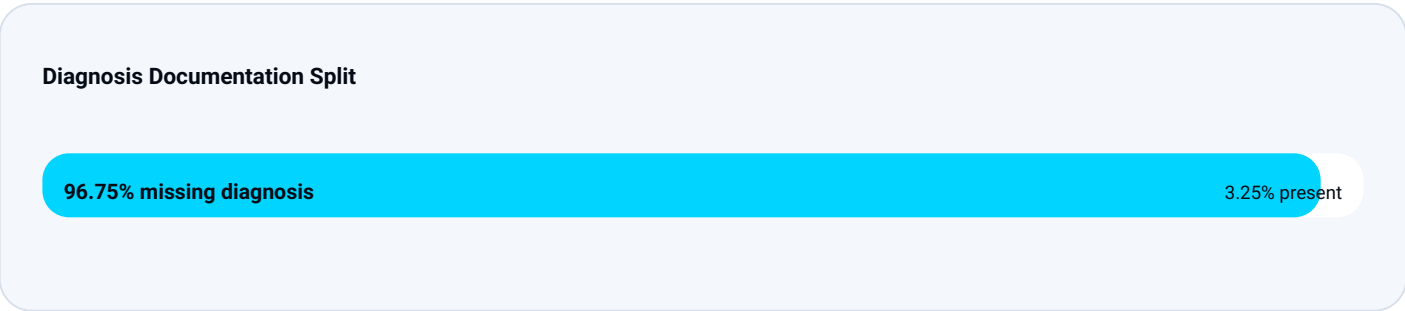
PRESCRIPTIONS REVIEWED

Systematic random sampling, March-May 2019.

100%

PRESCRIBER NAME MISSING

A linked documentation and accountability failure.



Clinical interpretation: when the diagnosis is absent, every downstream actor - pharmacist, emergency physician, follow-up doctor, caregiver - must infer intent from medicine names, dose patterns, or patient memory. IntelliScript BD defines this failure mode as Clinical Blindness: the prescription exists, but the clinical context needed to prescribe safely is structurally absent.

FORENSIC IMPLICATION

A handwritten prescription can be physically present and still fail as a safety record. The missing diagnosis rate transforms the document from a medical communication artifact into a partial transaction receipt. A health identity layer must preserve diagnosis context, allergy context, current medications, and prescription provenance together.

Source note: Rahman MM, Munia AT, Sikdar KMYK & Sarkar MR. A cross-sectional study on current prescription trends and errors in outpatient department of a Bangladeshi secondary care district hospital. Perspectives in Clinical Research, 13(3), 161-167.

The Bogura Study: Lethal Interaction Density

Paul et al. documented medication-related problems in 200 handwritten hospital prescription orders in Bogura, Bangladesh.

366

DDIS IDENTIFIED

Across 200 prescriptions.

12.57%

SERIOUS DDIS

Potentially severe/lethal interaction class.

63

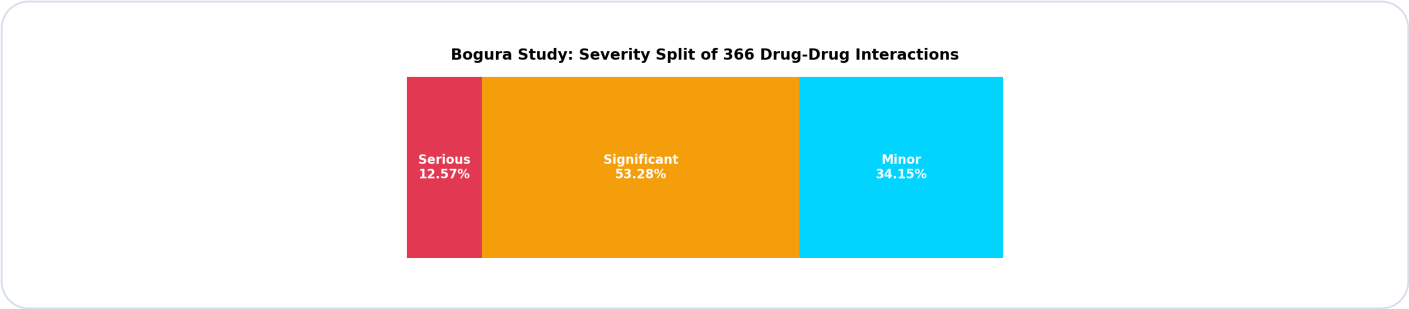
UNCLEAR DRUG NAMES

Ambiguous handwriting.

219

DOSE GAPS

Dose strength missing.



MEDICATION-RELATED PROBLEMS

The paper reported approximately 692 medication-related problems, or 3.46 per prescription, from 1,234 prescribed drugs. That translates into repeated opportunities for harm before the medicine even reaches the patient.

ARCHITECTURE REQUIREMENT

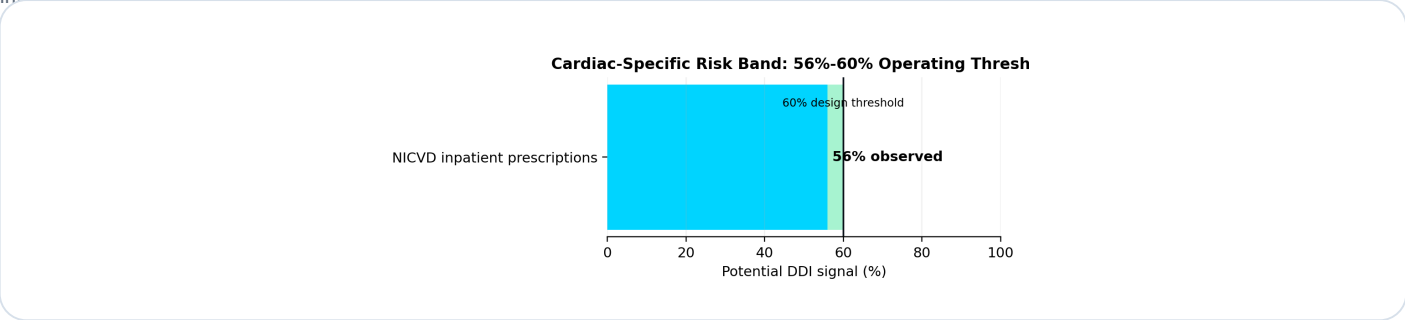
A viable system must not simply digitize a prescription image. It must parse medicines, validate dose strength, detect ambiguous drug identity, and block clinically dangerous combinations before dispensing.

Audit interpretation: in a paper-first workflow, prescription safety relies on legibility, memory, and manual review. The Bogura evidence proves that the hazard is not isolated to a single hospital desk or a single careless writer; it is a repeatable systems failure in handwritten order transmission.

Source note: Paul TR, Rahman MA, Biswas M, Rashid M & Islam MAU. Medication Errors in a Private Hospital of Bangladesh. Bangladesh Pharmaceutical Journal, 17(1), 32-37.

Cardiac-Specific Risk: 56%-60% DDI Operating Band

The cardiovascular setting is a high-risk stress test for prescription safety because polypharmacy, anticoagulants, PPIs, antibiotics, and comorbid drugs interact under time pressure



100

NICVD PRESCRIPTIONS

Randomly selected inpatients over three months.

56%

AT LEAST ONE DDI SIGNAL

Potential interacting drug combination observed.

14

HARMFUL INTERACTIONS

Potentially harmful pairs identified.

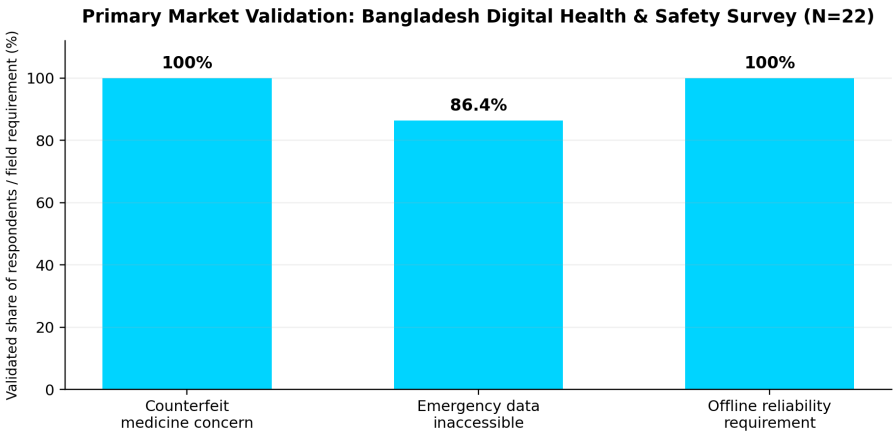
Critical Interaction Pairs Found in Cardiac Prescriptions		
Clopidogrel + Omeprazole	33.47%	Reduced antiplatelet activity; MI/death risk signal
Clopidogrel + Esomeprazole	27.75%	Reduced antiplatelet activity; hospitalization/death risk signal
Furosemide + Cephalosporin	10.62%	Increased nephrotoxicity risk
Atorvastatin + Vitamin B	7.76%	Myopathy/rhabdomyolysis risk signal
Warfarin + Moxifloxacin / Amoxicillin	1.1% each	Significant bleeding risk signal

DESIGN CONSEQUENCE

In cardiac and emergency contexts, a safe product must fail closed: when an interaction is severe, the prescribing interface must stop, explain, and require a deliberate override or medication change. This is the logic behind IntelliScript BD Sentinel Override.

Primary Market Validation: The N=22 Signal is Uniform

The Bangladesh Digital Health & Safety Survey validates patient-side urgency around counterfeit medicine, emergency data blindness, and digital health-passport adoption.



22/22

COUNTERFEIT CONCERN

All respondents were very or somewhat worried about fake medicine.

19/22

BLIND ALLERGY EXPOSURE

No instant access or only if family retrieved paper files.

22/22

DIGITAL-PASSPORT PULL

Yes or maybe to secure QR health passport.

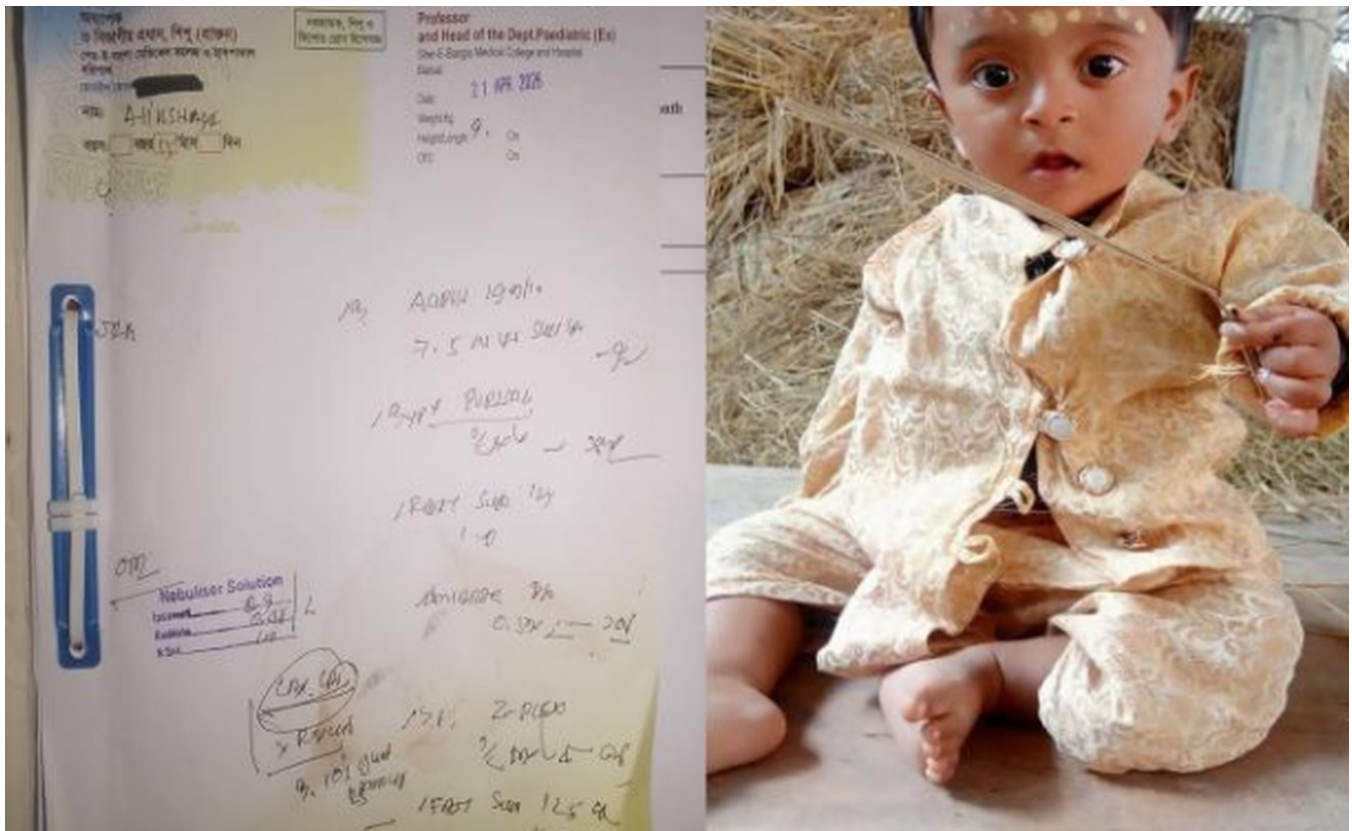
SURVEY METHODOLOGY NOTE

Verified worksheet contains 22 responses, timestamped 03 May-06 May 2026. The counterfeit metric aggregates “Very worried” and “Somewhat worried” responses. The emergency data metric aggregates “No idea” and “Only if family went home to find physical files.” Offline reliability is a field-coded requirement of the intended operating environment; the survey adoption proxy is 22/22 yes/maybe to a secure digital health passport.

Market interpretation: every respondent expressed counterfeit-medication concern, almost nine in ten lacked instant emergency data access, and every respondent showed willingness to engage with a sovereign QR health passport. That combination is unusually strong for a safety infrastructure product.

EV-001: Evidence of Clinical Blindness

The handwritten prescription hazard photo anchors the problem in real-world ambiguity: illegible orders, fragmented patient records, and downstream pharmacy interpretation risk.



EV-001 | HANDWRITTEN PRESCRIPTION HAZARD

Evidence of Clinical Blindness: a prescription image and patient-side documentation show how paper instructions can travel without a reliable machine-readable identity, allergy context, or interaction screen.

01

HANDWRITTEN ORDER

Medicine identity depends on human interpretation.

02

MISSING CONTEXT

Diagnosis/allergy data rarely travels with the script.

03

DISPENSING RISK

Pharmacist carries liability without a safety record.

04

NO AUDIT TRAIL

Patient cannot prove provenance or safety checks.

The Hustle: Patient Empathy Mapping

Founder Tabassum conducted frontline discovery in hospital waiting areas to map how prescriptions, phones, reports, and memory currently substitute for a trusted health identity layer.



Founder-led patient empathy mapping in a hospital waiting area. Focus: how patients carry, lose, explain, and reinterpret medical history across visits.

Follow-up patient interview showing the paper-and-phone hybrid reality: histories exist, but not in a structured, clinician-ready record.

FIELD SYNTHESIS

Patients did not describe a lack of documents; they described a lack of retrievable, trusted, clinician-readable context. The observed workflow is a fragmented archive: a phone photo here, a paper bag there, a family member’s memory, and a pharmacy interpretation. IntelliScript BD converts that archive into an authenticated health identity event.

The Hustle: Doctor and Pathology-Lab Validation

The field package documents founder consultations with medical and pathology stakeholders at Ahsania Mission Medical College, validating the problem across prescribing, triage, and lab-report workflows.



EV-D01

Consultation with a medical doctor at Ahsania Mission Medical College. Focus: prescribing burden, paper record gaps, allergy ambiguity, and emergency decision stress.



EV-L01

Consultation with pathology/lab staff. Focus: lab reports as disconnected artifacts and the need to bind observations to a durable patient identity.

CLINICAL WORKFLOW INSIGHT

Prescription safety cannot be solved at the pharmacy counter alone. It must begin where the order is generated and continue through lab data, allergy data, dispensing, and follow-up.

OPERATIONAL PROOF

The founders tested the problem in the exact environment where the product must operate: busy corridors, mixed paper/digital artifacts, low redundancy, and human time pressure.

Regulatory Pull Meets Daily Clinical Stress

Field notes and public legal evidence converge on the same conclusion: Bangladesh already recognizes prescription ambiguity as a patient-safety threat.

202

“Prescription Blindness is the #1 daily stressor for emergency surgeons.”

205

“Pharmacists are seeking a liability shield against Tk 13k-5 lakh DNCRP fines.”

2017 HIGH COURT MANDATE

Bangladesh High Court ordered steps for doctors to use capital letters or printed prescriptions so patients and dispensers can understand medicine names. The order directly maps to IntelliScript BD’s printed/QR prescription workflow.

DNCRP ENFORCEMENT REALITY

Public enforcement reporting shows DNCRP pharmacy actions ranging from local Tk 13,000 fines for expired medicines and irregularities to Tk 5 lakh counterfeit-medicine penalties. The liability signal is real.

Regulatory interpretation: IntelliScript BD is not merely improving UX. It operationalizes three legally and commercially relevant needs: legibility, provenance, and evidence-backed dispensing. A pharmacist receiving a QR-secured prescription with immutable payload and interaction/allergy warnings has a defensible workflow. A patient receives safer continuity. A doctor receives a structured, auditable record.

Evidence anchors: High Court legibility order coverage; Human Rights and Peace for Bangladesh petition materials; BSS DNCRP enforcement reports; Business Standard/BSS reports on counterfeit medicine fines.

Technical Moat: Offline-First Sovereign Health Identity

The product must survive Bangladesh's operating reality: intermittent power, low bandwidth, legacy devices, urgent admissions, paper carryover, and pharmacy liability.

SYS-QR

AD SECURED

IMMUTABLE RECORD

BROADCAST REFERENCE

RX-6BFE-5251



PATIENT DECRYPTIOKEY

449580

Immutable prescription payload with broadcast reference and patient decryption key.

SYS-AL

LOAD TRI

Active Clinical Session

Told Tawsof

ALLERGIES

ACTIVE CONDITIONS

PAST CONDITIONS

Active allergy banner: structured allergy state carried into clinical pad.

SYS-SO

SENTINEL OVERRIDE

PRESCRIPTION BLOCKED — CRITICAL ALERTS DETECTED

AI TOXICOLOGY ANALYSIS

Concomitant use of Ciprofloxacin (Adecin) and Azithromycin (Adiz) is contraindicated due to a severe risk of QT prolongation and Torsades de Pointes, a life-threatening ventricular arrhythmia. Both medications are known to prolong the QT interval. It is recommended to avoid co-administration and consider alternative antibiotic therapy.

DISMISS & MODIFY PRESCRIPTION

Sentinel Override blocks critical DDI patterns before prescription release.

Architecture Requirements Matrix	
100% offline persistence	Local-first clinical node persists prescription, allergy, and patient identity during load-shedding windows.
FHIR-compatible data model	Patient, MedicationRequest, AllergyIntolerance, Observation, and Encounter mapped for interoperability.
SHA-256 evidence trail	Payload hashing provides tamper-evidence and immutable broadcast reference for prescription provenance.
Legacy hardware ready	Browser-based interface and lightweight secure node support low-spec clinics and pharmacy terminals.
Fail-closed safety logic	Severe allergy/DDI screens must stop prescription release until modified or explicitly overridden.

WHY THIS IS A MOAT

A cloud-only prescription app fails where the crisis is most acute. IntelliScript BD's moat is not just AI; it is the combination of offline persistence, sovereign identity, standards-aligned clinical resources, QR provenance, and a safety engine designed to work before the network comes back.

Why IntelliScript BD is Architecturally Viable for the Region

The evidence shows a systemic gap, not a feature request. The solution must fit Bangladesh’s clinical, legal, infrastructural, and pharmacy realities simultaneously.

Regional Constraint	Failure Mode	IntelliScript BD Response
Power instability	Cloud-only systems fail during load-shedding and emergencies.	Local node keeps clinical session active and syncs later.
Paper prescriptions	Ambiguous handwriting and missing diagnosis cause blind dispensing.	Printed/QR prescriptions carry structured medicine + context.
Counterfeit fear	Patients cannot verify provenance at retail pharmacy counter.	Immutable payload, broadcast reference, and pharmacy verification flow.
Emergency admissions	New doctors lack allergy, medication, and blood-type context.	Sovereign health identity exposes emergency-safe minimum dataset.
Pharmacy liability	Dispensers face fines and reputational risk without audit trail.	Dispensing evidence trail, alerts, and tamper-proof record reduce liability.



Source Register: Formal Bibliography and Evidence Links

The following register consolidates peer-reviewed research, public legal evidence, regulatory enforcement reporting, and technical standards used in this dossier.

01. Rahman MM, Munia AT, Sikdar KMYK & Sarkar MR (2022). A cross-sectional study on current prescription trends and errors in outpatient department of a Bangladeshi secondary care district hospital. Perspectives in Clinical Research, 13(3), 161-167.

PMC Rahman: [pmc.ncbi.nlm.nih.gov/articles/PMC9345250/](https://pubmed.ncbi.nlm.nih.gov/articles/PMC9345250/)

02. Paul TR, Rahman MA, Biswas M, Rashid M & Islam MAU (2014). Medication Errors in a Private Hospital of Bangladesh. Bangladesh Pharmaceutical Journal, 17(1), 32-37.

BanglaJOL Paul: www.banglajol.info/index.php/BPJ/article/view/22311

03. Uddin MB, Nipa NY, Ahmed S, Haider B, Hasan SB & Yousuf A (2016). Possibility of Drug-Drug Interaction through Prescription Analysis at NICVD, Bangladesh.

NICVD DDI Study: www.pharmascigroup.us/articles/OJPP-2-104.php

04. Bangladesh High Court prescription legibility coverage: doctors must type prescriptions or write in block capitals; generic names also addressed.

High Court coverage: www.pakistantoday.com.pk/2017/01/10/bangladesh-court-orders-doctors-to-write-legibly

05. Human Rights and Peace for Bangladesh petition material on illegible prescription directions and printed/capital-letter prescriptions.

HRPB petition PDF: www.hrpb.org.bd/public/images/interest/Stop_writing%20illegible_Medical_Prescription_Dir

06. The Daily Star perspective on Bangladesh High Court recognition of illegible prescription risk and printed/capital-letter direction.

Daily Star: www.thedailystar.net/star-weekend/perspective/do-doctors-need-handwriting-classes-now-1358509

07. BSS News (2022). Four business establishments fined Tk 13,000 in Jhalakathi; DNCRP drive included pharmacies with expired medicine irregularities.

BSS Tk 13,000 fine: www.bssnews.net/district/83999

08. BSS News (2025). Lazz Pharma fined Tk 5 lakh in Khulna following counterfeit-medicine enforcement.

BSS Tk 5 lakh fine: www.bssnews.net/district/302908

09. HL7 International. FHIR standard - healthcare data exchange model referenced for Patient, MedicationRequest, AllergyIntolerance, Observation, and Encounter compatibility.

HL7 FHIR: hl7.org/fhir/

10. NIST FIPS 180-4 Secure Hash Standard - SHA-256 cryptographic hashing reference for tamper-evident payload fingerprinting.

NIST SHA-256: csrc.nist.gov/pubs/fips/180-4/upd1/final

11. Primary evidence file. The Bangladesh Digital Health & Safety Survey (Responses) (1).xlsx. N=22; worksheet timestamps 03 May-06 May 2026.

Internal survey workbook | internal://survey-workbook

12. Primary visual evidence package. Founder-supplied photographs and interface screenshots for IntelliScript BD problem validation, April-May 2026.

Internal field photos | internal://field-photo-package

CLOSING CERTIFICATION

This source register and the preceding pages constitute the official Evidence of Problem Research dossier for IntelliScript BD, prepared as a high-fidelity clinical audit package for the GRIC Startup League.